THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 8 — MATHS · NCERT-TYPE CHAPTER TEST SET 3

Chapter 4: **Quadrilaterals** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The diagonals of a rhombus: 1

(a) are always equal **(b)** bisect each other at right angles

(c) are parallel **(d)** do not meet

2. The angles of a quadrilateral are in the ratio 3:4:5:6. The largest angle is: 1

(a) 120° **(b)** 132° **(c)** 100° **(d)** 108°

3. In a parallelogram ABCD, $\angle A = (2x + 30)^\circ$ and $\angle C = (3x + 5)^\circ$. The value of x is: 1

(a) 30 **(b)** 26 **(c)** 25 **(d)** 24

4. In rhombus ABCD, $\angle ABC = 110^\circ$. The angle $\angle OBC$ (O is the point where the diagonals meet) is: 1

(a) 110° **(b)** 35° **(c)** 70° **(d)** 55°

5. The diagonals of a rectangle ABCD meet at O and $\angle AOB = 110^\circ$. Then $\angle OAB$ is: 1

(a) 70° **(b)** 55° **(c)** 110° **(d)** 35°

6. In a kite ABCD with AB = AD and CB = CD, $\angle B = \angle D = 110^\circ$ and $\angle A = 50^\circ$. Then $\angle C$ is: 1

(a) 110° **(b)** 90° **(c)** 200° **(d)** 70°

7. In an isosceles trapezium ABCD with AB $\parallel$ DC, $\angle A = 98^\circ$. Then $\angle D$ is: 1

(a) 82° **(b)** 262° **(c)** 41° **(d)** 98°

8. In which quadrilateral does exactly one diagonal bisect the other? 1

(a) kite **(b)** parallelogram **(c)** rhombus **(d)** rectangle

9. The angles of a quadrilateral are in the ratio 2:2:3:5. The largest angle is: 1

(a) 138° **(b)** 162° **(c)** 90° **(d)** 150°

10. In a parallelogram ABCD, $\angle A = (4x + 2)^\circ$ and $\angle C = (5x - 8)^\circ$. The value of x is: 1

(a) 15 **(b)** 9 **(c)** 10 **(d)** 11

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Can a quadrilateral have angles 120°, 100°, 80° and 60°? Give a reason. 2

12. The angles of a quadrilateral are in the ratio 1:2:4:5. Find the smallest angle. 2

13. In a parallelogram ABCD, $\angle A = (5x + 10)^\circ$ and $\angle B = (4x - 10)^\circ$. Find x and $\angle A$. 2

14. Can a quadrilateral have angles 95°, 85°, 90° and 100°? Give a reason. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. The angles of a quadrilateral are in the ratio 2:3:5:8. Find all the angles. 3

16. ABCD is a rhombus whose diagonals meet at O. If $\angle BAD = 76^\circ$, find $\angle OAB$, $\angle ABC$ and $\angle OBA$. 3

17. The diagonals of a rectangle ABCD meet at O and $\angle AOB = 110^\circ$. Find $\angle OAB$, $\angle OAD$ and $\angle AOD$. 3

OR

The angles of a quadrilateral are in the ratio 4:5:6:9. Find all the angles. 3

18. ABCD is a rhombus whose diagonals meet at O. If $\angle BAD = 94^\circ$, find $\angle OAB$, $\angle ABC$ and $\angle OBA$. 3

OR

The diagonals of a rectangle ABCD meet at O and $\angle AOB = 50^\circ$. Find $\angle OAB$, $\angle OAD$ and $\angle AOD$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. (i) Show that the sum of the angles of a quadrilateral is 360°. 5
(ii) Three angles of a quadrilateral are 84°, 96° and 112°. Find the fourth angle.

OR

(i) State two properties of the diagonals of a rhombus. 5
(ii) In rhombus ABCD the diagonals meet at O and $\angle BAD = 120^\circ$. Find $\angle OAB$, $\angle OBA$ and $\angle ABC$.

20. (i) Show that the sum of the angles of a quadrilateral is 360°. 5
(ii) Three angles of a quadrilateral are 105°, 65° and 98°. Find the fourth angle.

OR

(i) State two properties of the diagonals of a rhombus. 5
(ii) In rhombus ABCD the diagonals meet at O and $\angle BAD = 48^\circ$. Find $\angle OAB$, $\angle OBA$ and $\angle ABC$.

END OF QUESTION PAPER — Total: 40 Marks

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